



Postulated pipe rupture design load for dynamic analysis of feedwater system

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ABSTRACT

A transient analysis has been performed on the economizer line of the feedwater system of YGN 5&6 nuclear generating station subject to transient loading initiated by a postulated pipe rupture. During this transient, the thermal-hydraulic response was calculated using RELAP5/MOD3.1 code, and the hydrodynamic force time history was generated using REFORC and R5FORCE post-processors having a different force determination technique and their results were compared. This analysis has been performed to evaluate the suitability for calculating hydrodynamic force under such transient, the potential for numerical instabilities, and advantages and disadvantages of each post-processor.

1. INTRODUCTION

The design of safety-related nuclear piping systems requires the determination of the hydrodynamic force applied to the system following severe pipe rupture and steam/water hammer events. A transient analysis has been performed on the economizer line of feedwater system for YGN 5&6 nuclear generating station subject to transient loading initiated by a postulated pipe rupture just outside the main steam isolation valve room. During this transient, the thermal-hydraulic response on the economizer line was calculated using RELAP5/MOD3.1 code. And the hydrodynamic force time history was generated using REFORC and R5FORCE having a different force determination technique, and their results were compared. This pipe

rupture design load is used for input into structural analysis code to perform the dynamic analysis of piping systems.

This analysis has been performed to evaluate the suitability for calculating fluid induced force on the economizer line of feedwater piping undergoing postulated pipe rupture, and discuss the potential for numerical instabilities, and advantages and disadvantages of each post-processor.

2. PROBLEM DESCRIPTION

The feedwater system of YGN 5&6 supplies feedwater from deaerator storage tank to each of the two steam generators via feedwater nozzles. Redundant check valves are provided in each feedwater line to preclude blowdown of each steam generator following a postulated pipe rupture upstream of the check valves.

Figure 1 shows a piping isometric diagram of the economizer line of the feedwater system associated with steam generator 2 of YGN 5&6. The initial conditions of the economizer line are 1668.8 lb/hr in flow and 455 °F in temperature. The steam generator pressure is 1088 psia, the ambient pressure is 14.7 psia. Each check valve is fully open during normal plant operation. At simulated time of $t=2.0$ seconds, a completely circumferential pipe rupture with crack opening time of one millisecond is postulated at the welding joint just outside the main steam isolation valve room. After pipe rupture each check valve starts to close due to the pressure differences between upstream and downstream of valve, and is closed completely in 0.7 seconds. In simulation the system is allowed to stabilize for 2.0 seconds before the pipe rupture, and the simulation is allowed to continue to 4.0 seconds.

3. METHODS OF ANALYSIS

During the transient the thermal-hydraulic response was calculated using RELAP5/MOD3.1^[1] code. RELAP5 developed by EG&G Idaho, Inc. is a code designed to perform the thermal-hydraulic transient analysis of nuclear power plant and related systems. RELAP5 is based on a one-dimensional

two-fluid, six-equation model. The six-equation model is numerically solved using a semi-implicit finite difference scheme.

Various post-processors coupled with RELAP5 are available currently for hydrodynamic force time history calculation. The pipe rupture design load was generated using REFORC^[2] and R5FORCE^[3] which have a different force determination technique and their results were compared. REFORC computer program has implemented the momentum balance method, and R5FORCE computer program has implemented the force balance method for determining the fluid induced forces acting piping system. REFORC was originally developed by EDS Nuclear, Inc., and modified by S&L. R5FORCE was developed by EG&G Idaho, Inc.

4. RELAP5/MOD3.1 MODELING

Figure 2 shows the nodalization diagram of the RELAP5/MOD3.1 model from steam generator to just outside the main steam isolation valve room of the economizer line of feedwater system. The pipe network was divided into 146 volumes corresponding to a volume length of 1.5 times pipe diameter. The maximum time step size of 0.0002 was used in RELAP5 model.

The steam generator shown in Figure 1 was represented in the RELAP5 model by 'time dependent' control volume components. The check valves shown in Figure 1 were represented in the RELAP5 model by 'motor valve' components. Each check valve flow area was assumed to be linearly proportional to the stem position. The critical flow model was used at the each check valve discharge plane.

The pipe network shown in Figure 1 was modeled using RELAP5 'pipe' components for straight lengths of pipe and as 'single junction' components for elbows. These components permit the detailed geometric aspects of the piping network to be modeled. All pipe lengths, bends, and area change were accounted for, and appropriate dimensions, loss coefficients and piping roughness applied.

A break point at the end of the pipe shown in Figure 1 was represented in RELAP5 model by a 'branch' component, a 'motor valve' component and a 'time

dependent' control volume component representing the ambient. Under transient, the flow at this valve is generally choked and the critical flow model was used at the valve discharge plane. The upstream piping shown in Figure was represented in RELAP5 model by a 'motor valve' component and a 'time dependent' control volume component representing the upstream piping. Each valve was assumed to be opened and closed at the same time and at the same ratio to represent the pipe serving.

5. RESULTS AND DISCUSSION

All numerical results were obtained using RELAP5/MOD3.1 coupled with REFORC and R5FORCE post-processors. Right after the pipe rupture a rapid pressure drop was observed the move along the feedwater line. The pressure drop was attributed to the decompression wave, which was generated due to pipe rupture and the discharge of fluid into the atmosphere. The pipe rupture caused all check valves to rapidly close to prevent reverse flow from the steam generator. Due to the valve closure against the reverse flow, the largest pressure pulse was observed downstream of the 18" check valve. At the location of pipe rupture, the initial calculated peak pressure using RELAP5 was compared with analytical result using the Moody paper^[4]. The numerical result, 240 psi was lower than the expected result, 304 psi based on the Moody paper.

The force time history by REFORC and R5FORCE at each pipe segment indicated in Figure 2 was captured and compared. The trend of force time history was very similar to each other even though there were slight differences between R5FORCE and REFORC results. R5FORCE provided smoother results since it does not use the time derivative of mass flow in its calculation. Table 1 summarized the peak force results at each pipe segment in Figure 2..

The effects of the initial depressurization loads due to postulated pipe rupture and the loads due to the rapid closure of the check valves were clearly observed in each pipe segment. The piping between the check valves and the steam generator experienced both conditions of loading while the piping upstream of the check valves was essentially controlled by the decompressurization loads. These loadings are axial in nature, i.e., they act along the pipe axes

As an example, Figure 3, 4 and 5 show the typical force time histories by REFORC and R5FORCE for pipe segment F1, F5, and F7 (See Figure 2 for pipe segment designation) respectively. The effects of pipe rupture and rapid closure of the check valves were clearly indicated in the Figure 3 and 4. REFORC results was very similar to R5FORCE results in Figure 3 and 4. Because REFORC reads the fully opened valve area from RELAP5 and uses this value in the force calculations, the difference between R5FORCE and REFORC was observed during the check valve closure.

The slight difference in the initial peak force at the location of pipe rupture in Figure 5 was observed between REFORC and R5FORCE. We have attempted to apply the Moody paper to evaluate the initial peak force at the location of pipe rupture. For pipe segment F7 in Figure 5, the initial peak forces by REFORC and R5FORCE after pipe rupture was approximately -306.5 kips and -227.7 kips respectively. However the initial peak force by Moody paper was -228 kips for pipe segment F7. The R5FORCE result was comparable to the analytical result. R5FORCE eliminated instabilities and data spikes encountered with REFORC.

When the maximum time steps used by REFORC and R5FORCE are different, the differences in the forces were observed. REFORC used the RELAP5 restart/plot file to calculate its forces. On the contrary, R5FORCE used the r5-r5f file which it writes automatically. R5FORCE was less affected by the time step selection, and provided a more accurate response to the transient. And also R5FORCE was easier to use than REFORC and did not require a post-processing such as PREPREF in REFORC.

6. CONCLUSIONS

The economizer line of the feedwater system for YGN 5&6 undergoing postulated pipe rupture has been studied using RELAP5/MOD3.1 coupled with REFORC and R5FORCE post-processors. The following conclusions were obtained in this study. However a comprehensive study is required to evaluate force time history effectively in conjunction with an experimental and analytical comparisons.

- 1) RELAP5/MOD3.1 code provided the useful thermal hydraulic response for the economizer of the feedwater system undergoing postulated pipe rupture.

- 2) REFORC and REFORC post-processors are suitable for calculating hydrodynamic forces in this transient and provided similar force time histories. R5FORCE provided smoother results since it does not use the time derivative of mass flow in its calculation.
- 3) R5FORCE predicted more accurate forces when compared to REFORC at the pipe rupture exit in this RELAP5 model.
- 4) The slight differences in the forces were obtained due to the different maximum time steps used by R5FORCE and REFORC. R5FORCE was less affected by the time step selection and easier to use than REFORC.

REFERENCES

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2. D.J. Dichurski, PREPREF-DEC : A Computer Program to Extract Information from RELAP5 Output and Restart/Plot Files, and REFORC-DEC : A Computer Program for Calculating Fluid Forces Based on RELAP5-Results, S&L, 1994
3. J.C. Watkins, EGG-EAST-9232, R5FORCE/MOD3s : A Program to Compute Fluid Induced Forces Using Hydrodynamic Output From the RELAP5/MOD3.1 Code, EG&G Idaho, Inc., 1990
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Table 1 Comparison of Maximum and Minimum Forces
By REFORC and R5FORCE

Force No.	Maximum Force(lbf)			Minimum Force(lbf)		
	Time(sec)	By REFORC	By R5FORCE	Time(sec)	By REFORC	By R5FORCE
F1	2.0834	2.4591E3	2.4661E3	2.0442	-8.0717E3	-8.0494E3
F2	2.0834	4.2462E3	4.3537E3	2.0442	-1.3766E4	-1.3658E4
F3	2.0828	7.1587E3	7.0732E3	2.0438	-2.2153E4	-2.2115E4
F4	2.7100	5.5501E4	4.9707E4	2.0298	-1.3343E5	-1.2544E5
F5	2.7100	6.1488E4	6.0462E4	2.0194	-1.3920E5	-1.3923E5
F6	2.8550	2.0927E3	1.8711E3	2.0098	-3.4176E4	-3.4291E4
F7	4.0000	-3.3642E4	-4.0945E4	2.0010/ 2.0006	-3.0648E5	-2.2775E5
F8	2.0768	3.1918E3	3.1985E3	2.0514	-7.4706E3	-7.4501E3
F9	2.0766	5.5076E3	5.6142E3	2.0514	-1.2825E4	-1.2716E4
F10	2.0762	9.3046E3	9.3000E3	2.0512	-2.1262E4	-2.1233E4
F11	2.7150	2.0536E4	1.9173E4	2.0432	-5.8833E4	-6.0153E4
F12	2.0914	2.0000E4	2.0451E4	2.0350	-5.0172E4	-5.1172E4

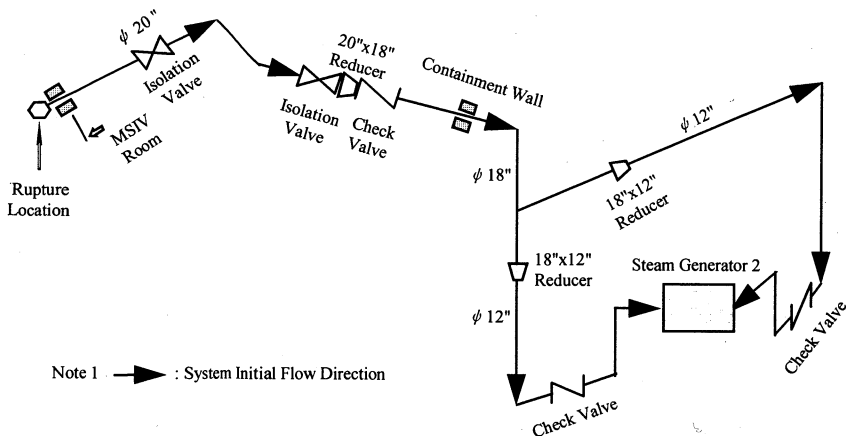


Figure 1 Piping Isometric Diagram of Economizer Line of Feedwater System for YGN 5&6

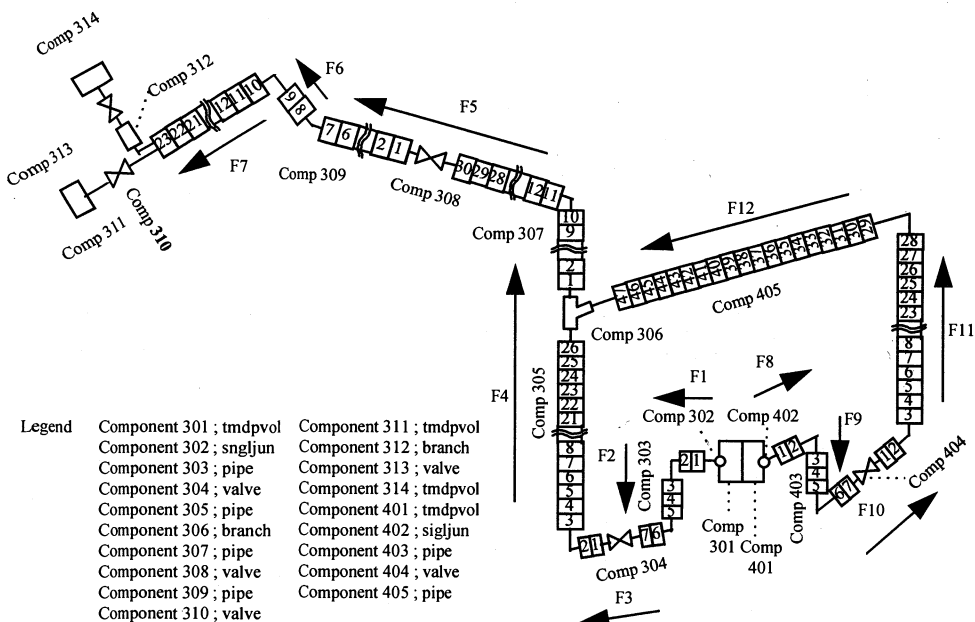


Figure 2 Nodalization Diagram of RELAP5/MOD3.1 Model

